# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2025

# MATHEMATICS

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Mathematics (Science) (Objective Type) - Group - I

**Time Allowed:** 20 Min. | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. $Cosec^2\theta - cot^2\theta$ = -------- :

(A) 0 (B) -1

(C) tan θ (D) 1

Q2. Value of x in 4 : x :: 5 : 10 is:

(A) 4 (B) 8

(C) 12 (D) 16

Q3. (A∪B)∪C = ------- :

(A) A∩(B∪C) (B) (A∪B)∩C

(C) A∪(B∪C) (D) A∩(B∩C)

Q4. The lengths of two transverse tangents to a pair of circle is:

(A) Over-lapping (B) Parallel

(C) Unequal (D) Equal

Q5. Identity $(5x + 4)^2 = 25x^2 + 40x + 16$ is true for:

(A) One value of x (B) Two values of x

(C) Three values of x (D) All values of x

Q6. $\alpha^2 + \beta^2$ = --------:

(A) $\alpha^2$-$\beta^2$ (B) $1/\alpha^2 + 1/\beta^2$

(C) $(\alpha + \beta)^2 - 2\alpha\beta$ (D) $\alpha + \beta$

Q7. The tangents drawn at the ends of diameter of circle are ---- to each other.

(A) Parallel (B) Non-parallel

(C) Collinear (D) Perpendicular

Q8. Two linear factors of $x^2 - 15x + 56$ are:

(A) (x + 7) & (x + 8) (B) (x - 7) & (x + 8)

(C) (x + 8) & (x + 7) (D) (x - 7) & (x - 8)

Q9. The third proportional of $x^2$ and $y^2$ is:

(A) $y^2/x^2$ (B) $x^2y^2$

(C) $y^4/x^2$ (D) $y^2/x^3$

Q10. The quadratic formula of $ax^2 + bx + c = 0$ is:

(A) $(b \pm \sqrt{(b^2-4ac)})/2a$ (B) $(-b \pm \sqrt{(b^2-4ac)})/2a$

(C) $(b \pm \sqrt{(b^2+4ac)})/2a$ (D) $(-b \pm \sqrt{(b^2+4ac)})/2a$

Q11. The number of elements in power set of {1, 2, 3} is:

(A) 8 (B) 6

(C) 4 (D) 2

Q12. An arc subtended a central angle of 40° then the corresponding chord will subtend central angle of --------:

(A) 20° (B) 40°

(C) 60° (D) 80°

Q13. The measure that determines the middle most observation in a data set is called:

(A) Median (B) Mode

(C) Mean (D) Geometric mean

Q14. The square roots of unity are:

(A) 1, -1 (B) 1, ω

(C) 1, -ω (D) ω, $\omega^2$

Q15. A complete circle is divided into:

(A) 90° (B) 360°

(C) 270° (D) 180°

## Mathematics (Science) (Subjective) (PART - I) - Group - I

**Time Allowed:** 2.10 hrs. | **Maximum Marks:** 60

Q2. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define with an example, the exponential equation.

(ii) Solve by factorization: $x^2 - x - 20 = 0$

(iii) Write the quadratic equation in standard form: $x/(x+1) + (x+1)/x = 6$

(iv) Find the nature of roots of equation: $x^2 - 23x + 120 = 0$

(v) Evaluate: $((-1+\sqrt{-3})/2)^9 + ((-1-\sqrt{-3})/2)^9$

(vi) Find the sum and product of roots of quadratic equation $(\ell + m)x^2 + (m + n)x + n - \ell = 0$

(vii) What is meant by joint variation?

(viii) If the ratios $3x + 1 : 6 + 4x$ and $2 : 5$, are equal find the value of x.

(ix) Find the third proportional to: $a^3 : 3a^2$

Q3. (Marks 12) Write short answers to any SIX (6) questions:

(i) If $(7x - 25)/((x - 4)(x - 3)) = A/(x - 4) + B/(x - 3)$, then find A and B.

(ii) Define improper fraction and give example.

(iii) If X = {2,4,6, ----, 20} and Y = {4,8,12,----,24} then find X-Y.

(iv) Define an onto function.

(v) If B={-1, 3} then find $B \times B$.

(vi) Write De Morgan's laws.

(vii) Find arithmetic mean by direct method: 12, 14, 17, 20, 24, 29, 35, 45

(viii) Define geometric mean.

(ix) Find the median of the given data: 2.9, 2.7, 2.5, 1.9, 2.3, 3.1

Q4. (Marks 12) Write short answers to any SIX (56) questions:

(i) Locate the angle -120°.

(ii) Find ℓ, when $\theta = 60° 30'$ , r = 15mm.

(iii) Convert into single trigonometric function: $\sin^2 x \cot^2 x$

(iv) Convert $3\pi/4$ radians to degree.

(v) Define segment of circle.

(vi) Differentiate between minor arc and major arc of a circle.

(vii) Define central angle of circle.

(viii) Construct a circle of radius 2 cm.

(ix) Define inscribed circle.

## Mathematics (Science) (Subjective) (PART - II) - Group - I

**Time Allowed:** 2.10 hrs. | **Maximum Marks:** 60

Note: Attempt THREE questions in all. But question No. 9 is Compulsory.

Q5(a). (Marks 4) Solve the equation: $2x + 5 = \sqrt{(7x + 16)}$

Q5(b). (Marks 4) Find the value of 'h' using synthetic division if '1' is the zero of the polynomial $x^3 - 2hx^2 + 11$

Q6(a). (Marks 4) Using theorem of componendo-dividendo, find the value of $(x - 3y)/(x + 3y) - (x + 3z)/(x - 3z)$ if $x = 3yz/(y - z)$

Q6(b). (Marks 4) Resolve into partial fractions: $(x^2 + 1)/(x^3 + 1)$

Q7(a). (Marks 4) If U = {1, 2, 3, ----, 10}, A={1, 3, 5, 7, 9}, B = {2, 3, 4, 5, 8} then prove that $(A \cup B)' = A' \cap B'$

Q7(b). (Marks 4) Compare the variation about mean for the two groups: A: 60, 70, 30, 90, 80, 40; B: 62, 62, 65, 68, 67, 48

Q8(a). (Marks 4) Verify the identity: $(1+ \sin\theta)/(1- \sin\theta) - (1- \sin\theta)/(1+ \sin\theta) = 4 \tan\theta \sec\theta$

Q8(b). (Marks 4) Inscribe a square in a given circle of radius 3 cm.

Q9. (Marks 8) Prove that two chords of a circle which are equidistant from the centre, are congruent. OR Prove that any two angles in the same segment of a circle are equal.

# GROUP – II

## Mathematics (Science) (Objective Type) - Group - II

**Time Allowed:** 20 Min. | **Maximum Marks:** 15

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. $\sec^2\theta$ = ------ :

(A) $1+ \tan^2\theta$ (B) $1- \sin^2\theta$

(C) $1+ \cos^2\theta$ (D) $1- \tan^2\theta$

Q2. In a ratio a: b, a is called:

(A) Relation (B) Consequent

(C) Antecedent (D) Proportion

Q3. The number of elements in power set {1, 2, 3} is:

(A) 4 (B) 6

(C) 8 (D) 9

Q4. Angle inscribed in a semicircle is:

(A) $\pi/2$ (B) $\pi/3$

(C) $\pi/4$ (D) $\pi/6$

Q5. $(2x + 1)/((x + 1)(x - 1))$ is --- :

(A) A proper fraction (B) An equation

(C) An improper fraction (D) A constant term

Q6. Cube roots of -1 are:

(A) $1,-\omega,-\omega^2$ (B) $-1,-\omega,\omega^2$

(C) $-1,\omega,-\omega^2$ (D) $-1,\omega,-\omega^2$

Q7. A line which has only one point in common with a circle is called:

(A) Sine of a circle (B) Cosine of a circle

(C) Tangent of a circle (D) Secant of a circle

Q8. The number of methods to solve a quadratic equation is:

(A) 3 (B) 2

(C) 1 (D) 4

Q9. Find x in proportion 4 : x :: 5 : 15 :

(A) 75/4 (B) 12

(C) 4/3 (D) 3/4

Q10. The solution set of equation $4x^2 - 36 = 0$ is:

(A) {3} (B) {±3}

(C) {4} (D) {±4}

Q11. A set with no element is called:

(A) Subset (B) Empty set

(C) Singleton set (D) Super set

Q12. A pair of chords of a circle subtending two congruent central angles is:

(A) Incongruent (B) Parallel

(C) Over-lapping (D) Congruent

Q13. A frequency polygon is a many sided:

(A) Triangles (B) Rectangle

(C) Square (D) Closed figure

Q14. If $b^2 - 4ac < 0$, then the roots of $ax^2 + bx + c = 0$ are:

(A) Rational (B) Real

(C) Irrational (D) Imaginary

Q15. The distance of any point of the circle to its centre is called:

(A) Diameter (B) Radius

(C) A chord (D) An arc

## Mathematics (Science) (Subjective) (PART - I) - Group - II

**Time Allowed:** 2.10 hrs. | **Maximum Marks:** 60

Q2. (Marks 12) Write short answers to any SIX (6) questions:

(i) Solve by factorization: $5x^2 = 15x$

(ii) Solve: $\sqrt{(3x + 18)} = x$

(iii) Define reciprocal equation and write one example.

(iv) Find the nature of roots : $x^2 - 23x + 120 = 0$

(v) Prove that the sum of the all cube roots of unity is zero.

(vi) Evaluate : $(1- \omega + \omega^2)^6$

(vii) Define proportion and give one example.

(viii) Find x if $6 : x :: 3 : 5$

(ix) If x and $y^2$ varies directly and $x = 27$, $y = 4$, find the value of k.

Q3. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define rational fraction and write example.

(ii) Resolve into proper fraction: $(3x^2 - 2x - 1)/(x^2 - x + 1)$

(iii) Write De Morgan's law.

(iv) If A={1,3,5,7}, B={2,3,5,7} then prove commutative property of union.

(v) If U = {1, 2, 3, ----10}, A = {2, 3, 5, 7} and B = {3, 5, 8}, find A' and B'.

(vi) Find the domain and range of R={(a,b), (b,a), (c,d), (d,e)}

(vii) Define standard deviation.

(viii) Write formula for finding the mode of grouped data.

(ix) Find the range: 11500, 12400, 15000, 14500, 14800

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) A flag pole 17.9 m high casts a 7 meter shadow. Find the angles of elevation of the sun.

(ii) Convert -7π/8 radians to degree measure.

(iii) Find r when $\ell = 52$ cm and $\theta = 45°$.

(iv) Verify: $\sin^2\theta/\cos\theta + \cos\theta = \sec\theta$

(v) How would you explain the concept of zero dimension briefly?

(vi) Differentiate between a chord and an arc of a circle.

(vii) Define circum angle.

(viii) Define escribed circle.

(ix) The length of the side of a regular pentagon is 5 cm. What is its perimeter?

## Mathematics (Science) (Subjective) (PART - II) - Group - II

**Time Allowed:** 2.10 hrs. | **Maximum Marks:** 60

> Note: Attempt THREE questions in all. But question No. 9 is Compulsory.

Q5(a). (Marks 4) Solve the equation : 5^(1+x) + 5^(1-x) = 26

Q5(b). (Marks 4) The difference of a number and its reciprocal is 15/4. Find the number.

Q6(a). (Marks 4) Using theorem of componendo-dividendo, solve the equation: (√(x+3) + √(x-3))/(√(x+3) - √(x-3)) = 4/3

Q6(b). (Marks 4) Resolve into partial fractions: $(7x + 4)/((3x + 2)(x + 1)^2)$

Q7(a). (Marks 4) If U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A = {1,3,5,7,9}, B = {2,3,5,7} then prove that (A∩B)' = A' ∪ B'

Q7(b). (Marks 4) Find standard deviation for the data: 9, 3, 8, 8, 9, 8, 9, 18

Q8(a). (Marks 4) The point (-3, 4) lies on terminal side of θ. Find the quadrant of θ and all six trigonometric ratios.

Q8(b). (Marks 4) Circumscribe a circle with regard to a right angle triangle with sides 3 cm, 4 cm and 5 cm.

Q9. (Marks 8) Prove that one and only one circle can pass through three non-collinear points. OR Prove that the measure of a central angle of a minor arc of a circle, is double that of angle subtended by corresponding major arc.